

FridgeChef: Risk & Mitigation Report

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Executive Summary

This report outlines the primary technical, operational, and financial risks associated with the FridgeChef AI-powered recipe generation platform. It details the specific mitigation strategies implemented to ensure system reliability, cost-efficiency, and user safety [cite: 76, 81].

Technical Risk Assessment

Risk Category	Description	Mitigation Strategy
Ingredient Detection	Inaccurate detection via computer vision leading to incorrect recipe bases [cite: 145].	Implementation of a manual correction UI allowing users to audit and edit detected items before generation [cite: 146].
Hallucination	AI generating nonsensical or dangerous recipes using unavailable ingredients [cite: 149].	Utilization of structured prompt constraints (JSON mode) and a secondary validation layer to enforce ingredient consistency [cite: 150].
System Latency	API response times exceeding the target	Asynchronous processing in FastAPI and implementation of

Risk Category	Description	Mitigation Strategy
	threshold of 3 seconds [cite: 112].	caching for common ingredient combinations [cite: 148].

Operational & Financial Risks

Risk Category	Description	Mitigation Strategy
API Cost Scaling	High operational costs due to excessive LLM API calls [cite: 147].	Prompt token optimization and database caching of high-quality, frequently requested recipes [cite: 148].
Data Privacy	Potential exposure of user dietary preferences or personal information [cite: 115].	Strict JWT-based authentication and a zero-selling policy for user data to ensure regulatory compliance [cite: 113, 115].

Compliance & Reliability

The system is designed for **95% or higher uptime** [cite: 116]. Reliability is maintained through Dockerized deployment on AWS/GCP, allowing for automated scaling and recovery [cite: 114, 123]. Security is handled through secure hashed password storage and encrypted database connections [cite: 113].